Vibrational spectral prediction of transient species using machine learning potential energy surfaces

Belinta Naomi Simiyu[†], **Jonathan M. Smith**[★†]

[†]*Temple University*

[★]*Hylleraas Centre, University of Oslo*

Highly energized species explore unique portions of molecular potential energy surfaces and undergo novel reaction dynamics. We have developed machine learning potential energy surfaces of energized species and radicals to explore dynamics. Molecular dynamics on these surfaces can be used to simulate experimental spectra. Experimental spectra can be used to refine and train machine learning potentials. Criegee intermediates are highly energized intermediates formed following atmospheric ozonolysis of alkenes and we use this context as the first application of our methodology. We have trained machine learning potential energy surfaces (MLPES) from importance sampled points DFT (B3LYP/cc-pVTZ) *ab initio* molecular dynamics (AIMD).

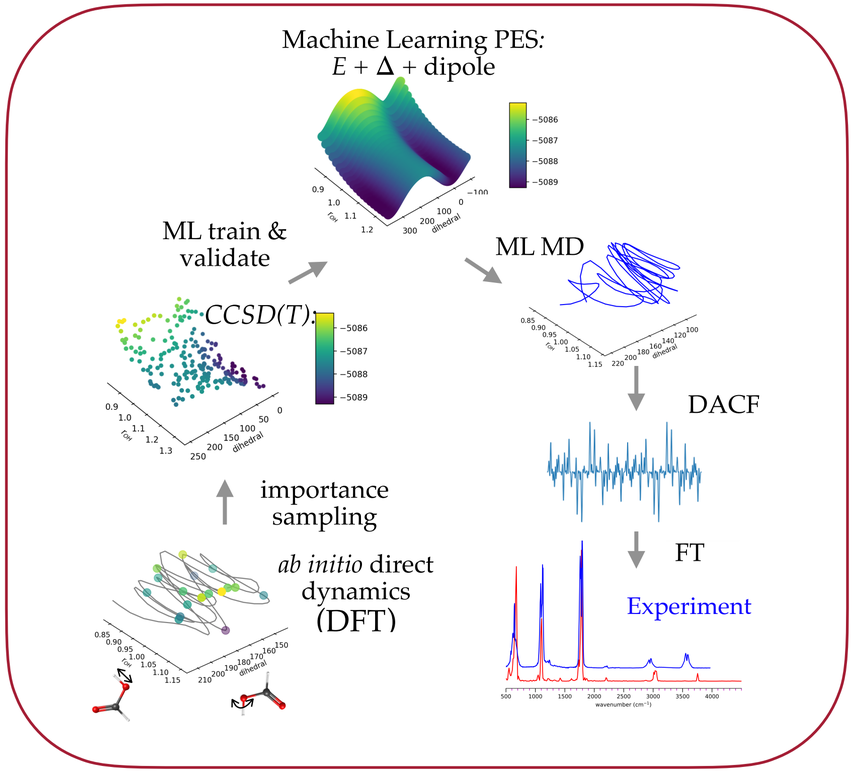


Figure 1. Overview of our machine learning potential energy surface construction protocol.

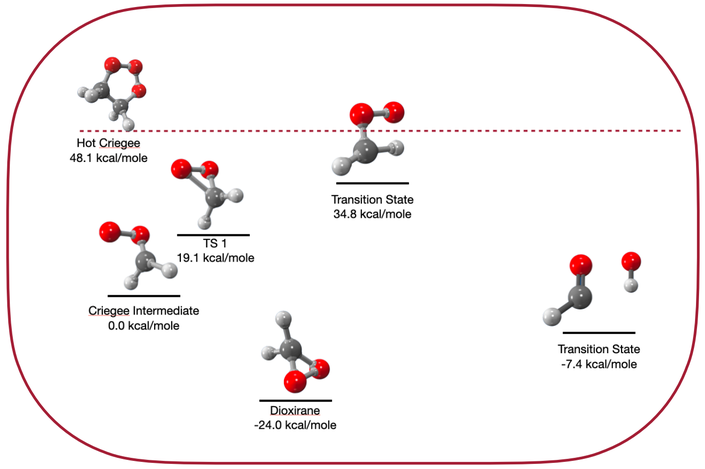


Figure 2. Example structures from one MLPES. Energetics using HEAT-345(Q) Protocol reported by Stanton *et al.*[4]

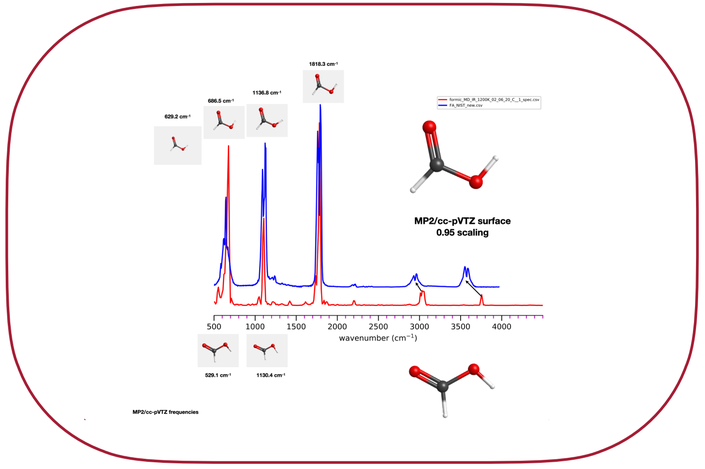
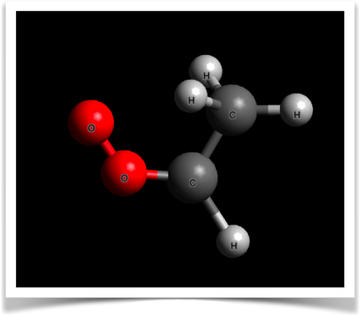


Figure 3. Example MLPES vibrational spectrum from Fourier transform of dipole autocorrelation function from molecular dynamics.



```
polynomial_dipole:
  Total RMSE: 0.043628 Debye
  Component RMSE (x,y,z): (0.027663, 0.018014, 0.028525) Debye
rbf_dipole:
  Total RMSE: 0.041281 Debye
  Component RMSE (x,y,z): (0.026444, 0.015431, 0.027690) Debye
gaussian_process_dipole:
  Total RMSE: 0.000898 Debye
  Component RMSE (x,y,z): (0.000610, 0.000271, 0.000601) Debye
```

Figure 4. Example dipole models.

References

(1) Smith, J. M.; Nikow, M.; Ma, J. Q.; Wilhelm, M. J.; Han, Y. C.; Sharma, A. R.; Bowman, J. M.; Dai, H. L. Chemical Activation through Super Energy Transfer Collisions. *Journal of the American Chemical Society* **2014**, *136* (5), 1682-1685.

(2) Smith, J. M. ; Nikow, M.; Wilhelm, M. J.; Dai, H. L. Collisional Relaxation of Highly Vibrationally Excited Acetylene Mediated by the Vinylidene Isomer. J Phys Chem A 2023, 127 (42), 8782-8793.

(3) Chmiela, S. Tkatchenko, A., Sauceda, H. E., Poltavsky, I., Schütt, K. T., Müller, K.-R., Science Advances, 3(5), 2017, e1603015

(4) Nguyen, T. L.; Lee, H.; Matthews, D. A.; McCarthy, M. C.; Stanton, J. F. Stabilization of the Simplest Criegee Intermediate from the Reaction between Ozone and Ethylene: A High-Level Quantum Chemical and Kinetic Analysis of Ozonolysis. J Phys Chem A 2015, 119 (22), 5524-5533



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